

TRACK 03 / ENTERPRISE INTELLIGENCE

KOHRA

Unified enterprise conversational AI

Grounded retrieval. Governed access. Traceable answers.

01 / THE PROBLEM

Disconnected policy knowledge creates friction and risk.

Employees lose hours navigating HR, Finance, Support, Privacy and Legal repositories. Unrestricted AI can expose confidential content, cite superseded policies and leave no compliance audit trail.

30

policy repositories

206

structured chunks

6

RBAC tiers

9

pipeline stages

02 / THE SOLUTION

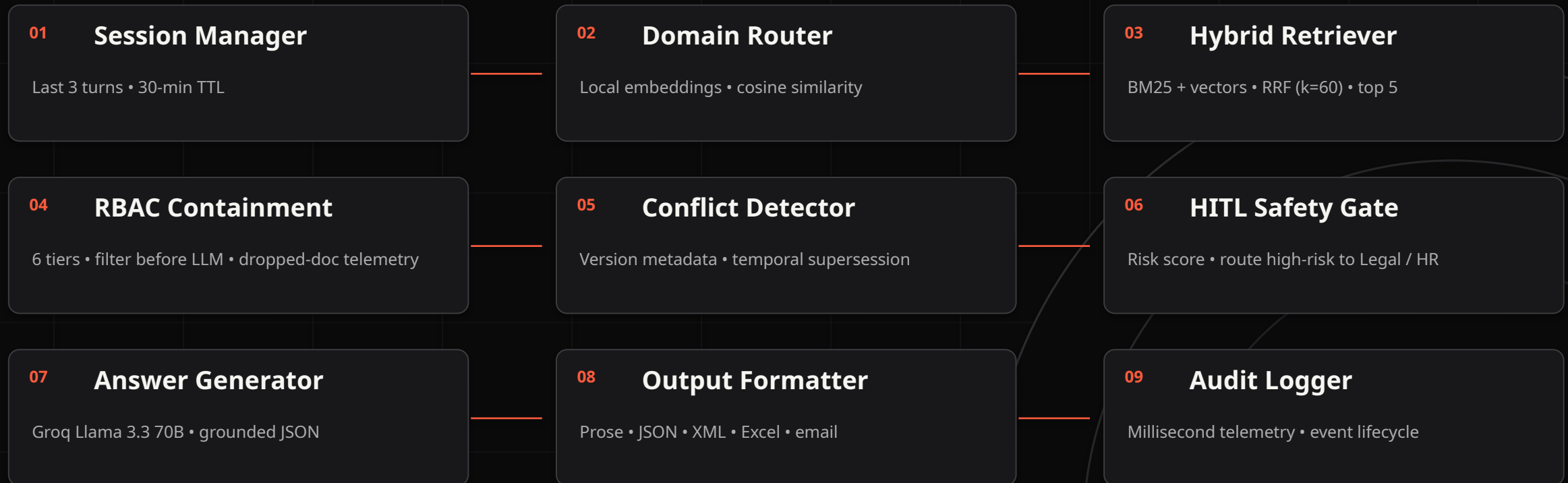
One agent. Six access tiers. Thirty verified repositories.

KOHRA combines hybrid retrieval, RBAC containment, supersession detection, human review for high-risk requests and grounded multi-format output across 206 structured chunks.

PIPELINE / RETRIEVAL TO RESPONSE

Nine stages. One governed answer.

React 19 → HTTP POST /api/chat → Express backend → Knowledge base



Knowledge base / 30 Markdown policies • BM25 inverted index • section chunk cache

BUILD / TECHNOLOGY AND DIFFERENTIATORS

Designed for grounded enterprise answers.

TECH STACK

FRONTEND	React 19 • TypeScript • Vite • Tailwind • Kohler Foundations
BACKEND	Node.js • Express • TypeScript • gray-matter • Handlebars • SheetJS
LLM	Groq Llama 3.3 70B
RETRIEVAL	BM25 inverted index • local semantic vectors • RRF
KNOWLEDGE	30 YAML-frontmatter Markdown files • 206 chunks
DEPLOYMENT	Local deployment • can be deployed

- 01 Hybrid RRF retrieval** Fuse BM25 keyword precision with semantic recall using Reciprocal Rank Fusion.
- 02 Post-retrieval RBAC** Filter unauthorized chunks before the LLM receives context; record dropped documents.
- 03 Temporal conflict detection** Use version metadata and date bounds to prefer current policy over superseded versions.
- 04 Human-in-the-loop gates** Risk-score sensitive requests and route exceptions to Legal or HR leadership.
- 05 Multi-format serialization** Render the same grounded answer as prose, JSON, XML, Excel or an email draft.

VALIDATION / DEMO HIGHLIGHTS

Five scenarios. Five passed.

Evaluation coverage across grounding, cross-domain synthesis, access control, versioning and output formats.

01	Single-domain grounding	Retrieved the \$75/day domestic travel per diem from Finance v2.1.
02	Cross-domain synthesis	Combined Privacy DPA and Legal MSA indemnification for a cloud-vendor query.
03	RBAC containment	EMPLOYEE access blocked 3 documents; FINANCE received executive-budget context.
04	Policy conflict resolution	Prioritized PTO v3.0 (20 days) over superseded v2.0 (15 days).
05	Multi-format output	Generated five formats, including a downloadable multi-sheet Excel file.

BUSINESS VALUE

~50%

faster time-to-answer

100%

policy traceability

0

ungrounded-source allowance

6 tiers

access governance

Grounded answers • honest refusal • HITL escalation • compliance logging

Kohler Foundations alignment / Vermilion accents • architectural typography